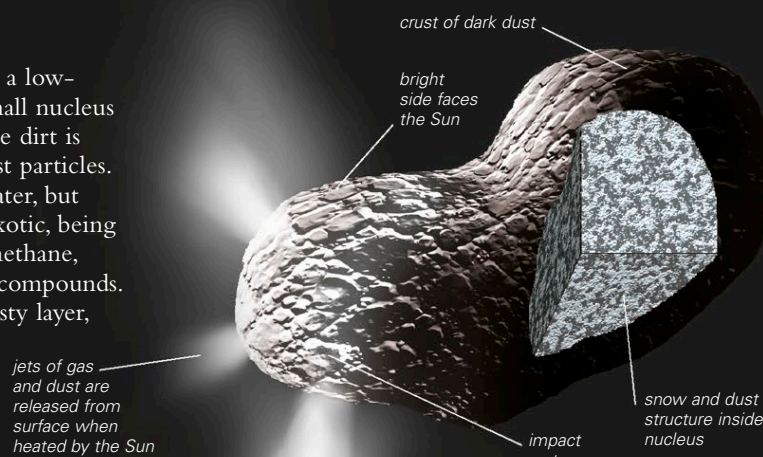


STRUCTURES

The fount of all cometary activity is a low-density, fragile, irregularly shaped, small nucleus that resembles a “dirty snowball.” The dirt is silicate rock in the form of small dust particles. The snow is mainly composed of water, but about 1 in 20 molecules are more exotic, being carbon dioxide, carbon monoxide, methane, ammonia, or more complex organic compounds. The nucleus is covered by a thin, dusty layer, which is composed of cometary material that has lost snow from between its cracks and crevices. The snow is converted directly from the solid into the gaseous state by the high level of solar radiation the comet receives when it is close to the Sun.

NUCLEUS

The central part of Comet Borrelly's elongated nucleus has a smooth terrain, but the more “mottled” regions consist of steep-sided hills that are separated by pits and troughs.

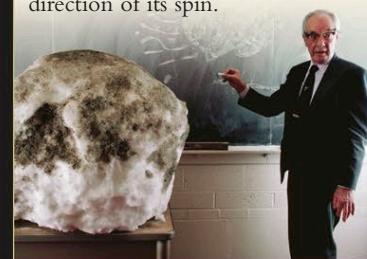


CROSS-SECTION

The nucleus has a uniform structure consisting of many smaller “dirty snowballs.” The surface dust layer is only a few inches thick and appears dark because it reflects little light. The strength of the whole structure is negligible. Not only do tidal forces pull comets apart, but many simply fragment at random.

FRED WHIPPLE

Fred Whipple (1906–2004) was an astronomy professor at Harvard University and the director of the Smithsonian Astrophysical Observatory from 1955 to 1973. In 1951, he introduced the “dirty snowball” model of the cometary nucleus, in which the snowball spins. As the Sun heated one side, its heat was slowly transmitted down to the underlying snows, which eventually turn straight to gas. This resulted in a jet force along the cometary orbit which either accelerated or decelerated the nucleus depending on the direction of its spin.

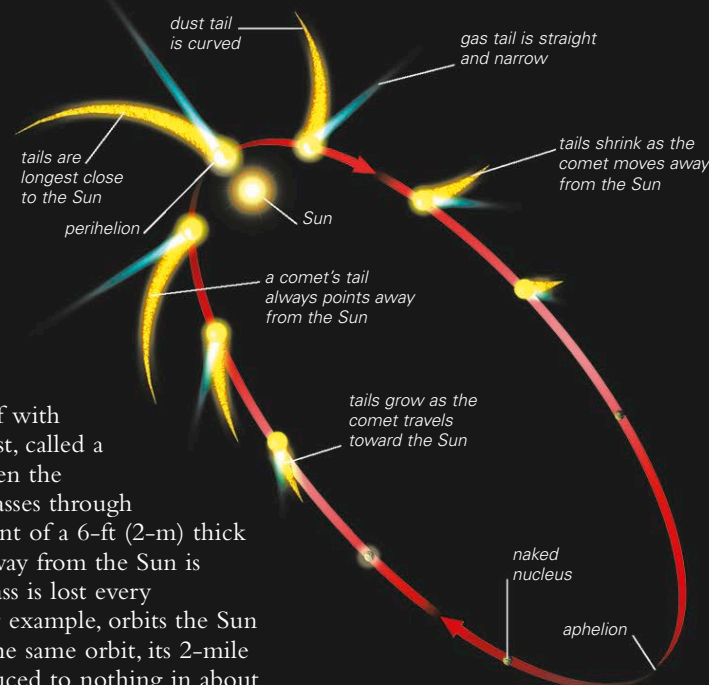


CRATER CHAIN

This 120-mile (200-km) long chain of impact craters, named Enki Catena, is on Ganymede, the largest of Jupiter's moons. It is likely that Ganymede was struck by 12 or so fragments of a comet that had just been pulled apart by tidal forces as it passed too close to Jupiter.

LIFE CYCLES

A comet spends the vast majority of its life in a dormant, deep-freeze state. Activity is triggered by an increase in temperature. When the comet gets closer to the Sun than the outer part of the Main Belt (see p.170), frozen carbon dioxide and carbon monoxide in the nucleus start to sublime—that is, they pass directly from the solid to the gaseous state. Once the comet is inside the orbit of Mars, it is hot enough for water to join in the activity. The nucleus quickly surrounds itself with an expanding spherical cloud of gas and dust, called a coma. The coma is at its maximum size when the comet is closest to the Sun. A comet that passes through the inner solar system will lose the equivalent of a 6-ft (2-m) thick layer from its surface. The comet moving away from the Sun is thus smaller than it was on its approach. Mass is lost every time a comet passes perihelion. Borrelly, for example, orbits the Sun about every 7 years. If it stays on the same orbit, its 2-mile (3.2-km) wide nucleus will be reduced to nothing in about 6,000 years. Comets are transient members of the inner solar system. They are soon dissipated by solar radiation. Large cometary dust particles form a meteoroid stream around the orbit. Gas molecules and small particles of dust are just blown away from the Sun and join the galactic disk.

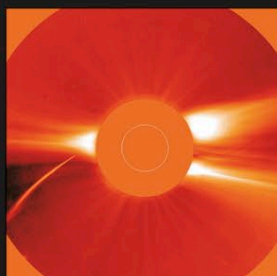


COMETARY TAILS

As a comet nears the Sun, it develops two tails. The curved tail is formed of dust that is pushed away by solar radiation. The straight tail consists of ionized gas that has been blown away from the coma by the solar wind.

COMET SOHO-6

Large numbers of sungrazing comets have been discovered by the SOHO satellite. Here, Soho-6 is seen as an orange streak, at left, approaching the masked Sun.



HALLEY'S TAIL

These 14 images of Halley's Comet were taken between April 26 and June 11, 1910, around the time it passed perihelion. An impressive tail was produced and dissipated in just 7 weeks of its 76-year orbit.